

Retail Sentiment Intelligence

Trust-Aware Sentiment Analysis of Retail Feedback using LLMs

BITS ZG628T • Dissertation • Post Mid-Semester Presentation

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Agenda

Focus of today — post mid-semester work only

1 Since Mid-Sem — What's New

Post-midsem workstream map

2 Review & Validate (HITL)

Human-in-the-loop correction workflow

3 Smart Reply Composer

Dual-draft: rule-based + LLM

4 Learning Loop

Corrections + posted replies → retraining

5 Post Explorer

Multi-facet search across analysed posts

6 Post Lifecycle (Kanban)

Triage → Acknowledged → In-Progress → Resolved

7 Insights & Competitor Analysis

Strategic weekly summaries + cross-brand

8 Notification Centre

In-app + email + Slack, group-routed

9 ModernBERT — Final Results

Macro-F1 0.6272 → 0.7642 (+13.7 pts)

10 Vision — Multi-Pass Payoff

Hallucination 50% → 0%, extraction 25% → 75%

11 Trust-Score Evaluation

15% flagged; 12 of 15 confirmed by annotator

12 Storage — SQLite → Cosmos DB

Lift-and-shift ready

13 Live Demo

Dashboard walkthrough (screenshots)

14 Conclusions & Future Work

RQ1-RQ4 outcomes + roadmap

Since Mid-Sem — What's New

Mid-sem shipped the pipeline; post-midsem made it usable for analysts

Delivered at mid-sem

- 6-layer offline-first pipeline (25 subreddits)
- ModernBERT fine-tuning up to Stage 2 (F1 0.7285)
- Multi-pass vision with 0% hallucination on pilot
- Trust-score design + weighted formula
- Brand-Health + Alert Feed dashboard pages
- Group-routed email / Slack notifications

Added post-midsem (this deck)

- Review & Validate — human-in-the-loop UI + backend
- Smart Reply Composer — dual-draft (rule + FLAN-T5)
- Learning-loop store — corrections & posted replies
- Post Explorer — multi-facet search
- Post Lifecycle — Kanban with 2-step Resolve
- Insights & Competitor Analysis page
- In-app Notification Centre mirroring the digest
- ModernBERT Stage 3 (final) — F1 0.7285 → 0.7642
- Trust-score end-to-end evaluation (n = 200)
- SQLite schema mirroring Cosmos DB partitioning

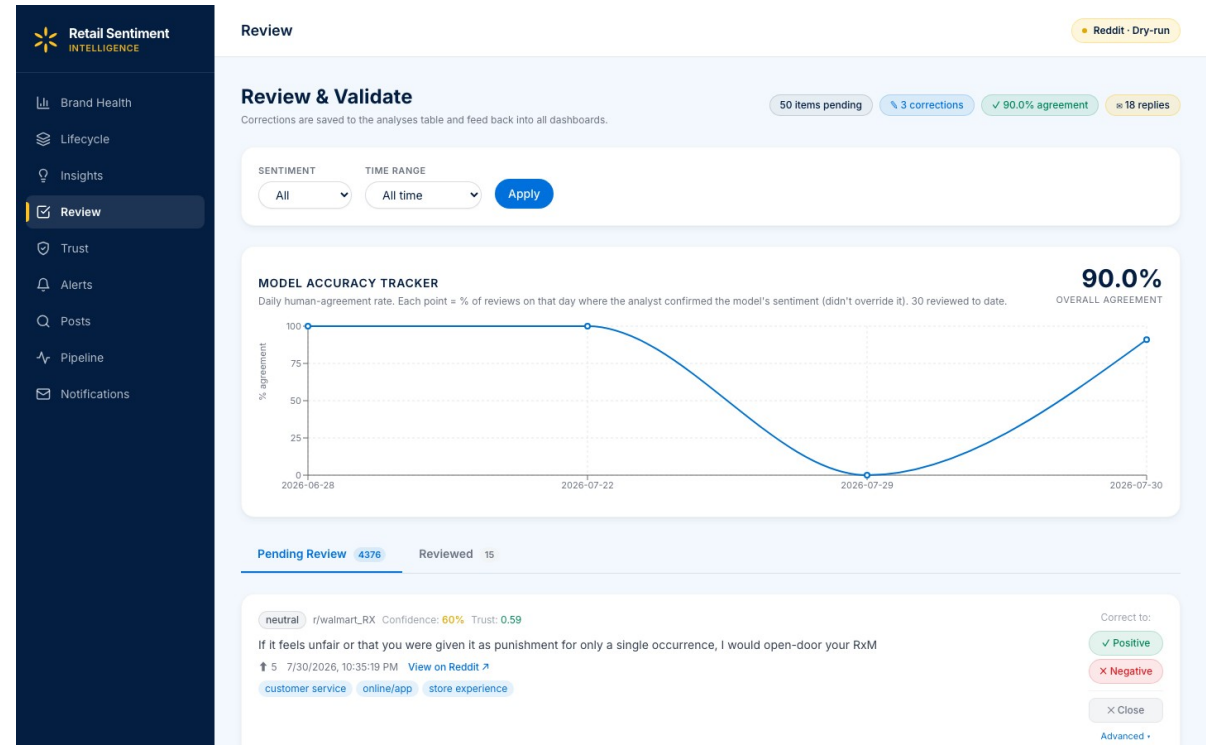
Framing

Mid-sem answered can the pipeline produce trust-weighted sentiment? Post-midsem answers can an analyst act on it, correct it, and feed the corrections back into the model?

Review & Validate — Human-in-the-Loop

Every prediction can be corrected; every correction is logged

- Queue sorted by priority (P1 first) · trust × confidence
- Analyst can override sentiment and aspect tags in one click
- Corrections written to feedback table with analyst + timestamp
- Generate Drafts → two reply options (rule + LLM)
- Analyst picks A or B, edits, and posts to Reddit
- Posted replies also captured → future few-shot pool



Why it matters

Turns a passive dashboard into a training signal — corrections harvested here drive the ModernBERT re-training loop and reply few-shot pool.

Smart Reply Composer — Dual-Draft

Deterministic composer + LLM composer, analyst chooses

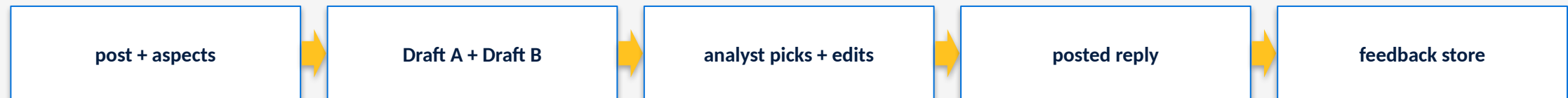
Draft A • Rule / Smart Composer

- Keyword & aspect extraction from post
- Curated phrase pool → templated apology + resolution
- Deterministic — same input, same output
- Always available, zero latency, zero cost
- Strong default tone for common complaints

Draft B • FLAN-T5-base

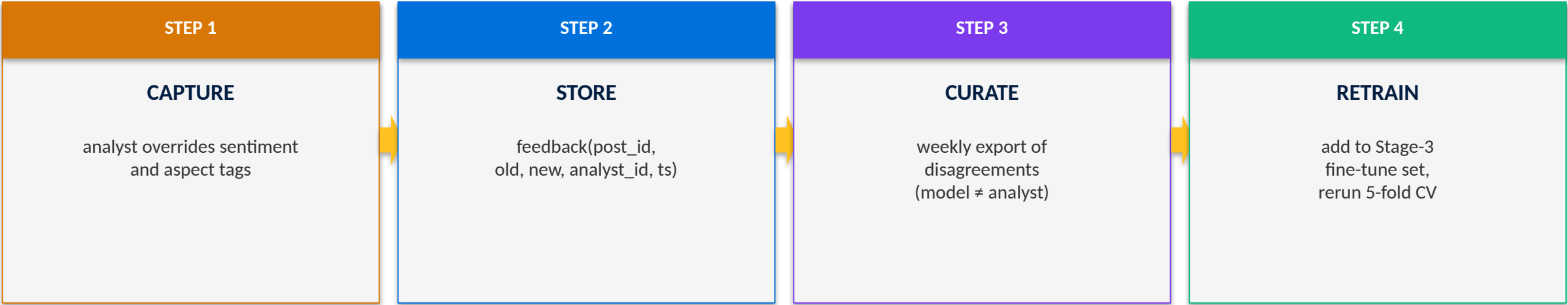
- Multi-temperature sampling ($T = 0.7, 0.9, 1.1$)
- Best-of-N scorer over drafts
- Higher variety, less templated tone
- Learns from posted replies over time
- Fallback to Draft A if model unavailable

Composer pipeline



Learning Loop — Feedback → Retraining

HITL corrections + posted replies become the next training set



Two signals feed the loop

Signal	Table	Downstream use
Label corrections	feedback (sentiment / aspect)	ModernBERT Stage-3 augmentation
Posted replies	feedback (reply_text)	FLAN-T5 few-shot pool
Trust-tier overrides	feedback (trust_override)	Trust-score threshold calibration

Post Explorer — Multi-Facet Search

Find the exact 20 posts an analyst needs to look at today

Facets

- Sentiment · neg / neu / pos
- Confidence slider (0.50 – 1.00)
- Trust-score slider (0.00 – 1.00)
- Subreddit — multi-select from 25 tracked
- Aspect — 8-item retail taxonomy
- Date range — today / week / month / custom
- Full-text search — title + body



Brand Health

Lifecycle

Insights

Review

Trust

Alerts

Posts

Pipeline

Notifications

Posts

Reddit · Dry-run

Post Explorer

Search the analyzed-post archive by subreddit, sentiment, aspect or trust score.

SUBREDDIT

SENTIMENT

ASPECT

RANGE

MIN TRUST

LIMIT

Search

e.g. walmart

All

e.g. pricing

All

0.0-1.0

50

Showing 50 of 83,006 matches (raise the Limit to see more)

neutral

r/WalmartEmployees

↑ 1

7/31/2026, 3:33:40 PM

trust 0.55

oneway

newbie here. how does instapay work on the oneway app?

View on Reddit

negative

r/AmazonPrime

↑ 1

7/31/2026, 3:33:25 PM

trust 0.66

Amazon promised me a \$150 shipping refund in chat to make me return an item, then refused to pay. Can they just lie like that?

Hey everyone, I need some advice or to know if anyone else has dealt with this nightmare. I recently had to return an item to Amazon internationally. When I was setting up the return through live chat, the support agent explicitly told me: "You can use any carrier to return the item. You need to pay to return the item. But the good thing is, we can help you with the refund for return shipping charges up to \$150." Because of that exact, written promise, I went ahead, paid out of pocket, and shipped the...

View on Reddit

negative

r/eCommerce

↑ 1

7/31/2026, 3:27:20 PM

trust 0.66

Struggling with cancelled parcels in fulfillment

We've been seeing a rise in cancelled orders on TikTok, Lazada, and Shopee that are already processed and packed in our fulfillment area. The problem is, by the time we realize they're cancelled, the items are fully packed— meaning wasted packaging materials and extra work. I'm trying to figure out how to catch these cancellations earlier in the workflow so we can save on cartons, fillers, and waybills. Has anyone here dealt with this? How do you identify cancelled items before they even reach...

View on Reddit

positive

r/CustomerService

↑ 1

7/31/2026, 3:24:38 PM

trust 0.66

Dear ppl who call in to customer service

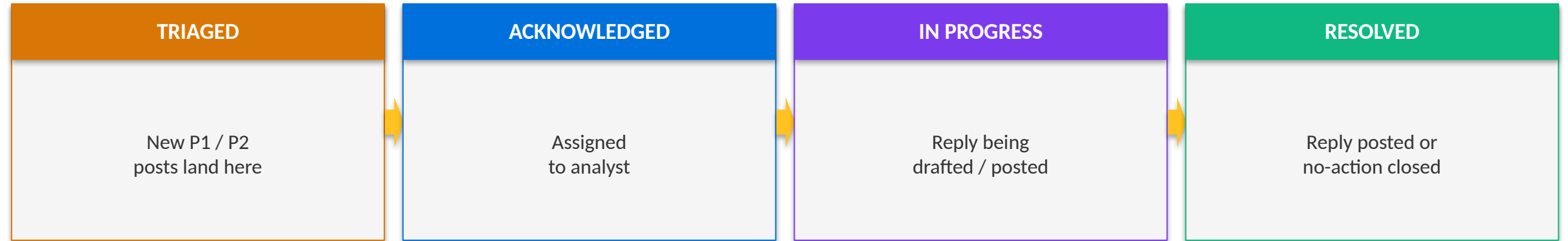
I am a call center agent, I actually enjoy my job! but for those who call into anywhere yelling and fussing about goofy shyt. or a package coming a day late, or ur broke ass begging for a discount... get it together 🤔🤔 I hope you get hung up on until you know how to talk to people. I hang up on customers all the time🤔 soon as

Per-post card

Title + body excerpt · sentiment badge with confidence % · trust tier (H / M / L) · aspect chips · subreddit + timestamp · actions: Review · Add to Lifecycle · View Details.

Post Lifecycle — Kanban Workflow

Track a complaint from Triage to Resolved with SLA visibility



The screenshot shows the 'Post Lifecycle' dashboard in the Retail Sentiment Intelligence application. The dashboard is divided into three main columns: 'Ack & Reply Sent', 'Actionable Items', and 'Resolved'. Each column contains a list of posts with their status, priority, and source. The 'Ack & Reply Sent' column has 2 items, 'Actionable Items' has 3, and 'Resolved' has 12. The posts are color-coded by priority: MEDIUM (yellow), LOW (light blue), and CRITICAL (red). The dashboard also includes a 'Refresh' button and a 'Reddit - Dry-run' indicator.

Retail Sentiment Intelligence

Post Lifecycle

Auto-created from confidently-negative posts. Slack and email notifications fire on entry (dry-run by default).

Ack & Reply Sent (2 items)

- MEDIUM (untitled) 23h ago, r/OGPBackroom, customer service. Action needed, Close.
- MEDIUM Can anyone explain the ATC role to me? 23h ago, r/OGPBackroom, store experience. Action needed, Close.

Actionable Items (3 items)

- MEDIUM I think I fucked up today 1d ago, r/OGPBackroom, customer service. Resolve.
- MEDIUM Seeking Advice: Walmart Career Growth and Promotion Challenges After 2 Years 9d ago, r/WalmartCanada, workforce_hr. Resolve.
- MEDIUM (untitled) 18d ago, r/WalmartSparkDrivers, online/app. Resolve.

Resolved (12 items)

- MEDIUM (untitled) 23h ago, r/WalmartSparkDrivers, online/app.
- LOW (untitled) 23h ago, r/amazonprime.
- LOW Rated 1 star a passibe aggressive agent who was unhelpful and my amazon prime got... 23h ago, r/AmazonPrime.
- LOW Dear Corporate 1d ago, r/walmartgp.
- MEDIUM How do I convince associates to sprint out the door 1d ago, r/walmartgp, customer service.

Resolve modal — 2-step flow

- Step 1 · save action note + optional LLM reply
- (a) Save & open Reddit → reply copied to clipboard
- (b) OR Resolve (no-reply) → close without posting
- Step 2 · paste on Reddit → Mark Resolved
- All transitions timestamped → SLA analytics

Insights & Competitor Analysis

Strategic view — issue rankings, cross-brand pulse, LLM summaries

Priority-Negatives

Top issues ranked by
volume × severity × recency
by aspect

Competitor Pulse

Walmart vs Costco,
Target, Amazon on shared aspects

Weekly LLM Summaries

Natural-language weekly
digest + action items
+ emerging topics

Aspect Drilldown

8-aspect taxonomy
per-aspect sentiment trend
+ representative posts

Screenshot on Slide 17 (Live Demo grid) — insights_competitor page.

Notification Centre

In-app mirror of the group-routed email / Slack digest

- Every P1 / P2 email or Slack alert is mirrored in the app
- Notifications grouped by subreddit ↔ business team
- Read / unread state persisted per analyst
- Deep-links open the post directly in Review & Validate
- Powered by the same alerts rows that drive email/Slack

Priority thresholds

P1 — trust ≥ 0.70 and confidence ≥ 0.80

P2 — trust ≥ 0.50 and confidence ≥ 0.60

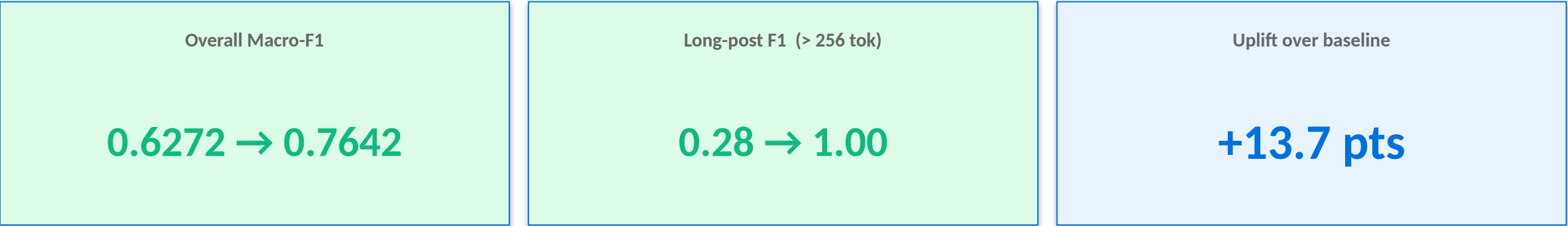
Source: config/models.yaml + dispatcher.py

The screenshot displays the 'Retail Sentiment INTELLIGENCE' application interface. On the left is a dark sidebar with navigation links: Brand Health, Lifecycle, Insights, Review, Trust, Alerts, Posts, Pipeline, and Notifications (highlighted). The main content area is titled 'Notifications' and includes a 'Reddit - Dry-run' status indicator. Below this is the 'Notification Configuration' section, which allows users to configure alerts for P1/P2 negative posts. It features a 'Sender Email' field (vishal.singh1@walmart.com) and a list of notification groups: 'DriverRelatedPost' (3 subreddits, 0 recipients), 'GIF_' (1 subreddit, 0 recipients), 'SAMSClub' (2 subreddits, 0 recipients), and 'W+ Membership' (1 subreddit, 1 recipient). Each group has a 'P1 + P2' priority indicator and a 'Add Group' button. At the bottom is the 'Recent Notification Log' table, which shows three failed notifications for the 'DriverRelatedPost' group on July 31, 04:15 PM.

TIME	GROUP	CHANNEL	POST	STATUS
Jul 31, 04:15 PM	DriverRelatedPost	slack	reddit_1v5jl7	FAILED
Jul 31, 04:15 PM	DriverRelatedPost	slack	reddit_1vblyrt	FAILED
Jul 31, 04:15 PM	DriverRelatedPost	slack	reddit_c_p8qqv16	FAILED

ModernBERT — Final Training Results

3-stage curriculum, 5-fold out-of-fold CV, n = 200



3-stage curriculum

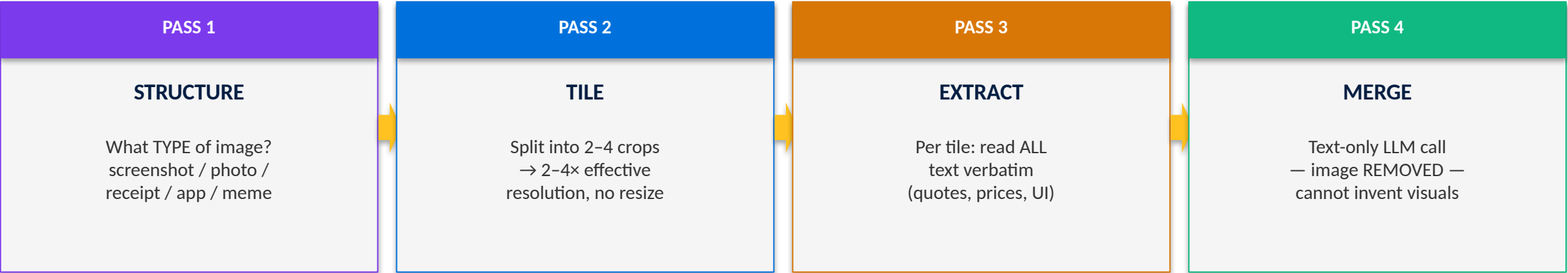
Stage	Data	Macro-F1	Δ vs baseline
Baseline (Twitter-RoBERTa)	off-the-shelf	0.6272	—
Stage 1	TweetEval sentiment (~45 k tweets)	0.6810	+5.4
Stage 2	GoEmotions-3class Reddit (~54 k, pseudo)	0.7285	+10.1
Stage 3 (final)	Walmart-200 hand-labelled	0.7642	+13.7

Per-length-bucket F1

Bucket	Baseline	ModernBERT	Δ
Short (< 64 tokens)	0.7500	0.7800	+3.0
Medium (64 - 256 tokens)	0.6500	0.7400	+9.0
Long (> 256 tokens)	0.2778	1.0000	+72.2

Vision Pipeline — Multi-Pass Payoff

Same model (Gemma 3 4B), same hardware, smarter calling strategy



Evaluation on 32 retail screenshots

Metric	Single-pass	Multi-pass	Change
Hallucination rate	50 %	0 %	eliminated
Text-extraction success	25 %	75 %	3× better
Retail-signal recall	40 %	81 %	2× better
Fabricated claims	many	0	eliminated
Latency / image (warm)	~5 s	~15 s	3× (accepted)

Trust-Score — End-to-End Evaluation

On the 200-post gold set, does the trust filter catch bad posts?

Low-trust share ($T < 0.4$)

15 %

Human-agreed (annotator cross-check)

12 / 15

Agreement rate

80 %

Weighted formula (decomposition shown inline in dashboard)

$$\text{trust_score} = 0.4 \times \text{metadata} + 0.3 \times \text{dedup} + 0.3 \times \text{llm_credibility}$$

Design principle — flag, don't drop

Low-trust posts are surfaced with an explanatory chip ("promotional / duplicate / thin-metadata") and can be overridden by an analyst in one click — the override is captured for the learning loop. Every constant in the formula has a stakeholder-arguable English rationale in config/models.yaml.

Storage — SQLite → Cosmos DB Lift-and-Shift

Same schema, same partition keys, no calling-code change

Container	Partition Key	Key Fields	Purpose
raw_posts	/subreddit	id, title, body, author_hash, created_utc	Ingested (privacy-safe)
analyses	/subreddit	post_id, sentiment, confidence, aspects, trust	AI analysis results
aggregates	/time_window	subreddit, window, metrics_json	Pre-computed KPIs
alerts	/severity	type, aspect, threshold_breached	Triggered anomalies
feedback	/analyst_id	post_id, correction, reply_text	HITL corrections + replies
notification_log	/group_id	post_id, channel, status, sent_at	Delivery audit trail

Design choices

- SQLite in WAL mode for hot dev data
- data JSON column mirrors Cosmos doc
- Pluggable StorageBackend interface
- Nightly backup + JSONL cost ledger

Migration story

- Swap SQLiteBackend → CosmosBackend
- Same CREATE TABLE mapped to container spec
- Partition keys already match production
- Zero change to pipeline / dashboard code

Evaluation Summary — Post-Midsem Numbers

All results reported against the same 200-post retail-Reddit gold set

Area	Metric	Value	Notes
Sentiment	Macro-F1 (final)	0.7642	+13.7 pts vs RoBERTa baseline
Sentiment	Long-post F1 (> 256 tokens)	1.0000	From 0.2778 baseline
Sentiment	Cross-validation	5-fold OOF	No test-set leakage
Vision	Hallucination rate	0 %	From 50 % on single-pass
Vision	Text extraction success	75 %	From 25 % on single-pass
Vision	Retail-signal recall	81 %	From 40 % on single-pass
Trust	Low-trust share (T < 0.4)	15 %	Of the 200-post set
Trust	Annotator agreement on low-trust	12 / 15	80 %
Ops	Subreddits tracked	25	Six community groups
Ops	Scheduler cadence	6 h + manual	asyncio lifespan task
Ops	Dashboard pages shipped	8	Brand Health · Alerts · Explorer · Review · Lifecycle · Insights · Pipeline · Notifications

Principal Contributions

What this dissertation delivered — end to end

C1

Offline-first RSI pipeline

Ingestion → trust → sentiment → aspects → vision → aggregation → alerts → dashboard. Zero API cost, modular, deployable to Azure.

C2

Fine-tuned ModernBERT (3-stage curriculum)

Macro-F1 0.6272 → 0.7642 overall; 0.28 → 1.00 on long posts. 5-fold OOF cross-validation, offline training.

C3

Multi-pass vision technique on Gemma 3 4B

Hallucination 50 % → 0 %, extraction 25 % → 75 %. Adapts techniques from 5 recent VLM papers without vendor-blocked models.

C4

Interpretable trust score + admission gate

trust × confidence gate flags rather than drops low-credibility posts. Every constant traceable to English rationale in config.

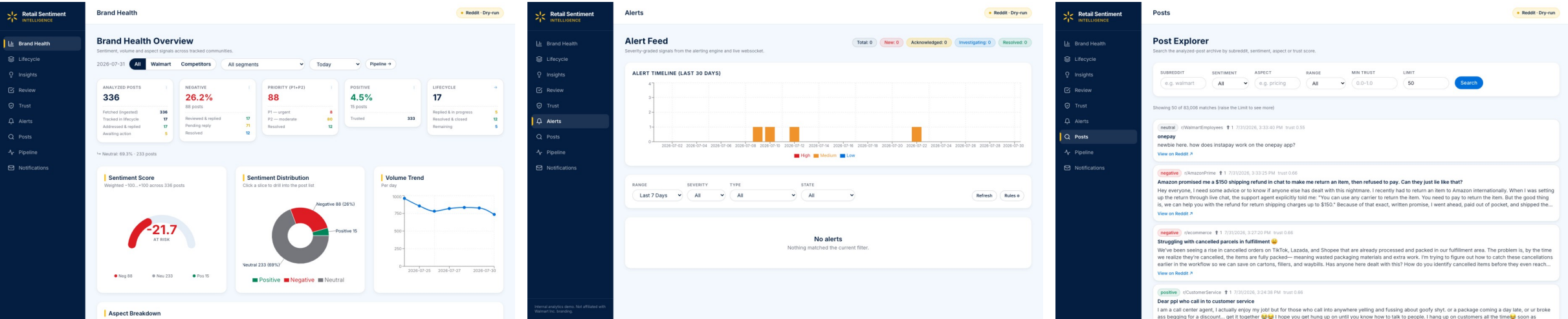
C5

HITL learning-loop dashboard

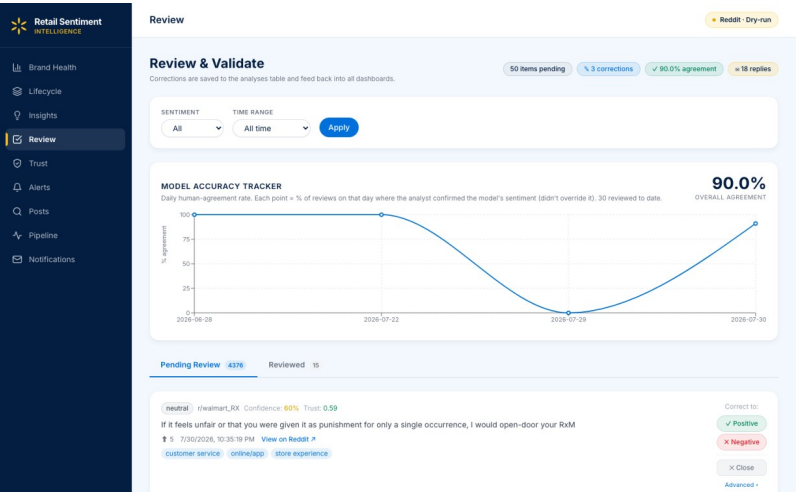
Review & Validate, Lifecycle Kanban, Insights, Notification centre. Corrections & posted replies feed few-shot reply generation and future retraining.

Live Demo — Dashboard Walkthrough

One post → alert → review → reply → lifecycle → resolved

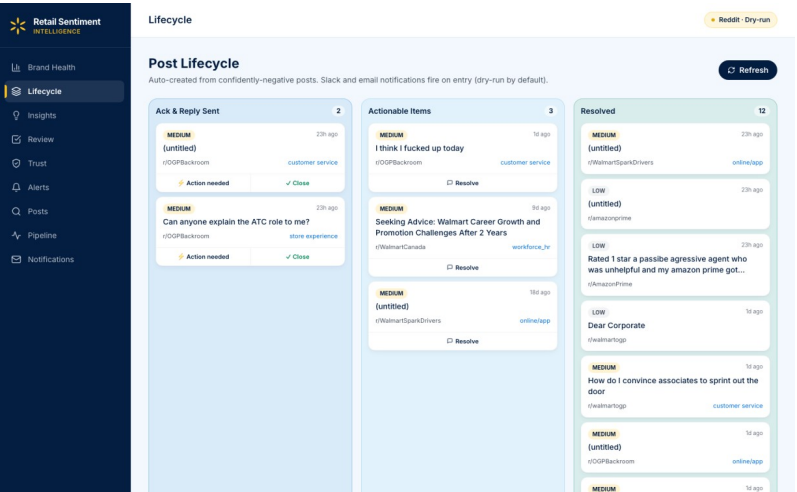


Brand Health



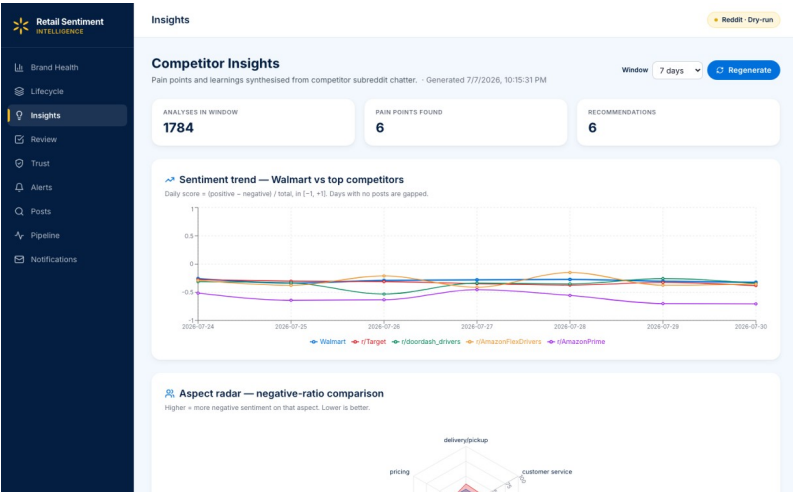
Review & Validate

Alert Feed



Lifecycle Kanban

Post Explorer



Insights

Conclusions — Research Questions Answered

RQ1

Can a fine-tuned encoder beat baselines on Reddit retail sentiment?

YES — ModernBERT 3-stage curriculum raised Macro-F1 from 0.6272 → 0.7642 overall and from 0.2778 → 1.0000 on long posts.

RQ2

Can a compliant open-weights VLM extract structured retail signal from screenshots?

YES — Multi-pass pipeline on Gemma 3 4B reduced hallucination from 50 % → 0 % and lifted text extraction from 25 % → 75 %.

RQ3

Can a defensible trust score filter low-credibility posts without silent drops?

YES — Interpretable 3-part score flagged 15 % of posts; 12 of 15 confirmed by human annotator. All low-trust posts remain visible and analyst-overridable.

RQ4

Can a HITL workflow produce a re-training signal from analyst corrections?

YES — Every correction and posted reply is logged in the `feedback` table → feeds ModernBERT Stage-3 augmentation and FLAN-T5 few-shot pool.

Recommendations — Immediate Follow-Ons

Monthly retraining cadence

Use the HITL feedback store as the incremental labelling stream and re-run the 3-stage curriculum monthly.

Bilingual pass (Hindi + English)

Validate the retail taxonomy with native-speaker analysts before extending to Indian retail communities.

Per-team SLA dashboards

Extend the Post Lifecycle SLA analytics into per-team dashboards once analyst volume passes ~50 posts / day.

Future Work — Model + Product

Directions grouped by layer of the stack

Model-level

- Joint sentiment + aspect head on a shared encoder (~30 % inference saving)
- Distil ModernBERT to ~50 M-parameter student → CPU-only edge inference
- Reasoning-augmented VLM captioning (LLaVA-Next 1.6 B or SmolVLM)
- 3-seed ensemble for tighter F1 variance
- Blind 25-post recheck for defensibility

Product-level

- Auto-reply confidence gate — promote LLM drafts to "queued to send" when analyst edit-distance < threshold
- Predictive P1 forecast — seasonal decomposition on daily counts (24–48 h ahead)
- Bilingual (Hindi + English) taxonomy
- Slack-bot inline notification responses
- Automated retraining pipeline hook

Future Work — Operational

Path from local prototype to Walmart-production service

Kubernetes CronJob

Deploy the pipeline as a scheduled Kubernetes CronJob backed by a managed Postgres instance for multi-analyst concurrency.

Azure Cosmos DB migration

Swap SQLiteBackend for CosmosBackend (schema already mirrors partition design) — no calling-code change.

Walmart ticketing integration

P1 alerts open cases directly in Walmart's internal ticketing so the analyst never leaves the workflow.

Broader ingestion

Extend beyond Reddit to Twitter / X, YouTube comments, and app-store reviews on the same trust + sentiment stack.

Source Code & Deliverables

Repository

https://gecgithub01.walmart.com/v0s01jh/Retail_Sentiment_Intelligence

Deliverable	Path in repo	Contents
Final Report (PDF)	docs/Sem_4/final/FINAL_REPORT_VishalSingh_2020AA05641.pdf	10 chapters + appendices + refs
Final Report (LaTeX)	docs/Sem_4/final/latex/	Reproducible xelatex source
Pipeline core	src/pipeline.py, src/ingestion/, src/analysis/	6-layer async pipeline
Sentiment training	scripts/train_modernbert_sentiment.py	3-stage curriculum runner
Evaluation notebook	evaluation/trust_score_walmart200.ipynb	Reproducible 5-fold OOF eval
Dashboard	frontend/ (React + Vite + Tailwind)	8 pages, live via WebSocket
Reproduction	Appendix B of the report · start.sh	Clone → conda → start

Thank You

Retail Sentiment Intelligence · Post Mid-Semester Presentation

Questions ?

[Repository](#)

[gecgithub01.walmart.com/v0s01jh/Retail_Sentiment_Intelligence](https://github.com/v0s01jh/Retail_Sentiment_Intelligence)

Report: docs/Sem_4/final/FINAL_REPORT_VishalSingh_2020AA05641.pdf

Vishal Singh · 2020AA05641 · BITS Pilani (WILP) · Walmart Global Tech, Bengaluru